

Vivek Ranjan

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PERSONAL SUMMARY

Senior Software Engineer with ~ 5 years of experience designing and building production-grade GenAI, Agentic AI, RAG, and cloud-native applications using Python, FastAPI, LangChain, LangGraph, and Cloud Services. Experienced in developing scalable LLM-powered APIs, AI guardrails, evaluation frameworks, multimodal AI, and enterprise data platforms. Built and scaled AWS data pipelines processing 10M+ patient records across various therapeutic areas while delivering AI-enabled solutions across enterprise pharmaceutical environments. **Eligible to work in Ireland without employer sponsorship (Stamp 1G)**

TECHNICAL SKILLS

Languages & Backend: C++, Python, FastAPI, REST APIs, SQL

Generative AI & Machine Learning: RAG, Agentic AI Workflows, LangChain, LangGraph, LangSmith, MCP, OpenAI API, Amazon Bedrock, Microsoft Foundry, Prompt Engineering, Multimodal AI, NLP, Vector DB, PyDantic, PyTorch

Cloud Platforms: AWS (Lambda, Glue, ECS, S3, IAM, VPC, Athena, Redshift, RDS, API Gateway, SageMaker, CloudFormation), Azure (Data Factory, Azure Databricks)

Data Engineering & Analytics: PySpark, ETL Pipeline Automation, Snowflake, Microsoft Fabric, Microsoft SQL Server, Apache Airflow, Apache Kafka, Delta Lake, Power BI, LAAD

DevOps & Collaboration: Docker, GitHub Actions, Jenkins, CI/CD, Git

EXPERIENCE

Eli Lilly and Company

November 2023 – August 2025

Associate Consultant

Bengaluru, India

- Engineered an intelligent analytics platform (BioSmartPlatform) using **LangChain, Python, and AWS** for Tempo (Diabetes), Lilly Health (Zepbound & Mounjaro), and Lilly Together (Immunology), leveraging **RAG** to retrieve product info from PDFs, key metrics from **Amazon Redshift and RDS** databases via Text-to-SQL and enterprise data sources via natural-language queries; automated data-quality and refresh monitoring, notifications, and **Jira** request management, saving the analytics team **8+ hours per week**
- Built high-concurrency **RESTful APIs with FastAPI, Python and LangGraph**, integrating LLM Agentic framework with scalable backend services for Consensus system (RedStrike) having **AI guardrail layers** including input/output validation, topic classification, and PII detection ensuring HIPPA/GDPR
- Built and scaled reporting pipelines for Zepbound (Lilly Health) & Diabetes (Tempo), processing **10M+** patient records across **150+** tables on AWS (S3, Glue, Redshift); reduced manual ops by **40%**
- Spearheaded the integration of SCD 1 and SCD 2 to enhance data accuracy and instituting a monitoring system for AWS applications, ensuring 99% uptime within 6 months
- Consolidated data in SQL Data Warehouse by connecting multiple tables to generate a unified flat table structure, and streamlining data reconciliation across sources, boosting accuracy and efficiency by 90%.

Dish Network Technologies (EchoStar Corporation)

September 2022 – October 2023

Software Engineer

Bengaluru, India

- Built automated **ETL workflows** using **PySpark** and **AWS Glue**, eliminating manual overhead and recovering **40+** engineering hours per month.
- Modernized **20+** warehouse and supply chain reports for El Paso and Denver operations using Python and SQL, improving reporting efficiency by approximately **20%**
- Built end-to-end ETL solutions using Azure Data Factory V2, automating data extraction, pipeline orchestration, and monitoring through Power BI
- Integrated ServiceNow with data pipelines to enable failure alerts, reducing support resolution time by 30%

Cognizant Technology Solutions

January 2021 – September 2022

Programmer Analyst

Remote

- Created a reliable CI/CD pipeline using GitHub Actions along with a flexible, config-driven Python framework, enabling the automated build, daily updates, and auditing of warehouse transactional tables
- Modelled star-schema fact and dimension tables for Commercial Insurance dataset to support scalable BI and analytics use cases, applying SQL, PySpark, and data warehousing best practices in Snowflake

VIBE CODING PROJECTS

GraphCore Studio | *Local-First Visual Agent Workflow Platform*

- Built a local-first visual AI workflow studio with drag-and-drop graph editing, conditional routing, agent handoffs, structured-output validation, Python tools, and human approval checkpoints
- Designed a native **C++17 graph execution engine** exposed to Python through a C ABI/**ctypes** bridge, supporting workflow scheduling, state management, and crash-recoverable checkpoints
- Integrated OpenRouter models, Pydantic validation, plugin-based Python tools, and a dependency-free browser interface for local workflow authoring and execution
- Published the project to PyPI with one-command installation via **pip install graphcore-studio**, using CMake and scikit-build-core for native runtime packaging
- **GitHub Repository:** github.com/vivek19398/graphcore-studio
PyPI Package: pypi.org/project/graphcore-studio/

AstraPDF | *Python PDF Library and Local Web Studio*

- Developed a cross-platform Python PDF toolkit and local browser studio for merging, splitting, extracting, rotating, reordering, annotating, watermarking, cropping, and exporting PDF documents
- Implemented a privacy-first architecture using a loopback-only web server and in-memory document processing, ensuring uploaded files remain local and are not transmitted to an online service
- Integrated native **PDFium** rendering with lazy thumbnail generation, page-image caching, and short-lived in-memory PDF sessions to improve large-document browsing performance
- Published the package to PyPI with one-command installation via **pip install astrapdf**; exposed a Python API through the **pdfstudio** module and a command-line launcher via **astrapdf**
- **GitHub Repository:** github.com/vivek19398/AstraPDF
PyPI Package: pypi.org/project/astrapdf/

EDUCATION

University of Limerick <i>Master of Engineering, Artificial Intelligence</i>	September 2025 - August 2026 <i>GPA: 3.55/4 (First Class Honours)</i>
Panjab University, India <i>Bachelor of Engineering, Computer Science and Engineering</i>	July 2017- May 2021 <i>GPA: 70%</i>

RECOGNITIONS & ACHIEVEMENTS

- **Eli Lilly Rise Award (Q1 2024)** – Teamwork & Innovation in data infrastructure
- **Eli Lilly Rise Award (Q4 2024)** – Exceptional productivity & delivery

CERTIFICATIONS

- **Microsoft Certified: Azure Fundamentals (AZ-900)** — Microsoft, 2022
- **IBM RAG and Agentic AI** — Coursera, 2026